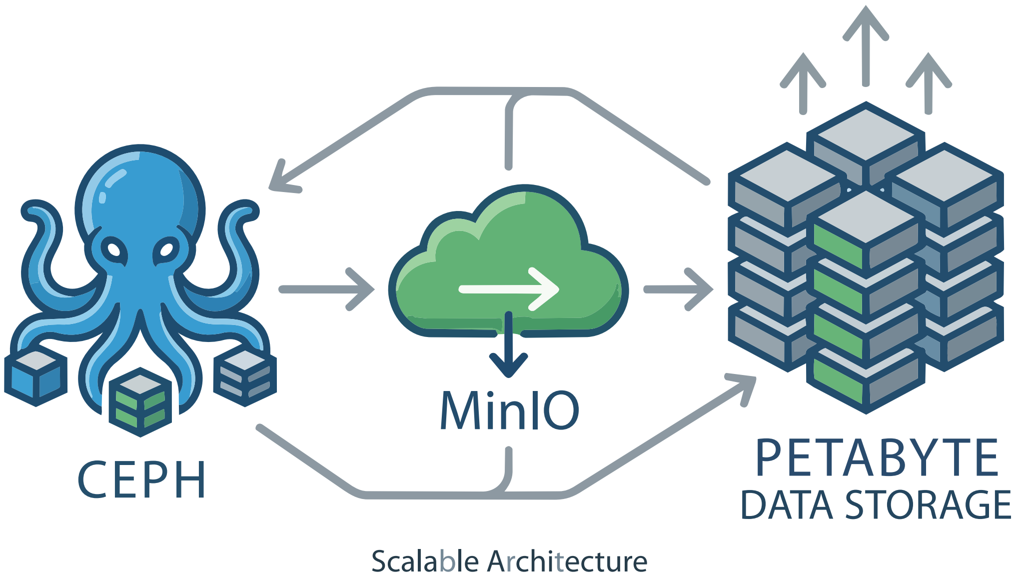
Using Ceph and MinIO for Scalable Petabyte Data Storage

Here's a practical guide to deploying, configuring, and operating open source object storage at scale.



Many organisations will run out of single-server storage before they run out of budget when using cloud storage to deal with large training sets, log archives, media libraries, or scientific datasets. Once you reach the petabyte scale, object storage is the way to go. Traditional file systems and block storage can be replaced by scalable flat namespaces with HTTP access.

Many people use MinIO and Ceph together. Ceph is good for large storage and MinIO for fast access. Ceph contains a lot of complex object storage features and can be used with its full set of features, or you can use MinIO with Ceph as a backend. MinIO can also be used with Ceph when you need a scalable storage system with complex features.

Ceph has many benefits:

- Block, object, and file storage don't require running separate systems.
- It is built to scale out and can run on commodity hardware.
- It automatically repairs by re-replicating the data.
- You avoid vendor lock-in. The fully open source (LGPL) model eliminates the ongoing per-TB costs associated with purchasing proprietary SANs or large scale cloud storage solutions.

The advantages of MinIO are:

- *S3 API compatibility:* Has drop-in replacement for AWS S3, which means existing tools (boto3, aws-cli, Spark, dbt, training frameworks) work unmodified.
- *Simplicity:* A single binary, minimal configuration, and fast time-to-first-bucket compared to Ceph's more involved cluster setup.
- *Performance:* Optimised for high-throughput read/write, which matters for GPU training jobs that are I/O-bound on dataset loading.
- *Erasures coding is built in:* MinIO handles data protection natively without needing a separate storage backend.

Both work well together -- Ceph is the long-lasting storage software, while MinIO instances can either be used as the S3 gateways or run on their own to help with training. The layout gives the scaling and long-lasting software of Ceph along with the S3 and simple operations software of MinIO along the edge.

The prerequisites for installing Ceph are:

- 3+ nodes recommended for production (minimum for quorum and replication); can be tested on a single host with cephadm in a lab configuration.
- Linux (Ubuntu 22.04/24.04 or RHEL/CentOS 9 recommended).